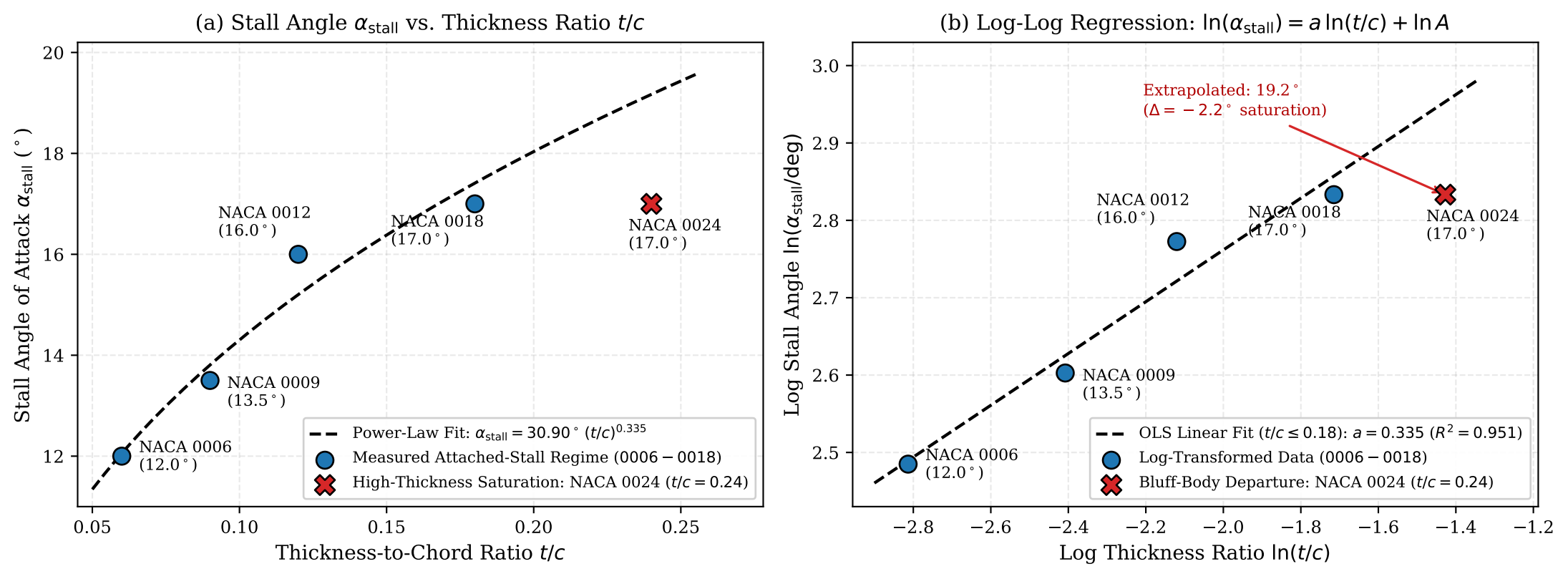
(a) Stall Angle α_{stall} vs. Thickness Ratio t/c
Stall Angle of Attack α_{stall} (°)
Thickness-to-Chord Ratio t/c
NACA 0012 (16.0°)
NACA 0018 (17.0°)
NACA 0024 (17.0°)
NACA 0009 (13.5°)
NACA 0006 (12.0°)
Power-Law Fit: $\alpha_{stall} = 30.90° \, (t/c)^{0.335}$
Measured Attached-Stall Regime (0006 − 0018)
High-Thickness Saturation: NACA 0024 ($t/c = 0.24$)
(b) Log-Log Regression: $\ln(\alpha_{stall}) = a \ln(t/c) + \ln A$
Log Stall Angle $\ln(\alpha_{stall}/\mathrm{deg})$
Log Thickness Ratio $\ln(t/c)$
Extrapolated: 19.2°
(Δ = −2.2° saturation)
NACA 0012 (16.0°)
NACA 0018 (17.0°)
NACA 0024 (17.0°)
NACA 0009 (13.5°)
NACA 0006 (12.0°)
OLS Linear Fit ($t/c \le 0.18$): $a = 0.335$ ($R^2 = 0.951$)
Log-Transformed Data (0006 − 0018)
Bluff-Body Departure: NACA 0024 ($t/c = 0.24$)